

DepthPhys: Near-Infrared Remote Photoplethysmography in Driver Monitoring Systems

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Abstract—Subtle variations in the body can be used to detect underlying physiological signals from camera video data. Existing research in this space is largely focused on extracting physiological signals from traditional two-dimensional planar video data. This project explores how the addition of depth data may be used to augment the results of these approaches. In particular, a novel optical pathway is proposed for camera-based physiological sensing and is validated on a new dataset consisting of seven hours of three-dimensional volumetric near-infrared monochromatic video data. Both two-dimensional and three-dimensional signals are extracted from this dataset and the results of either approach are compared using state-of-the-art camera-based physiological sensing neural methods. These findings provide valuable and novel insight into the utility of three-dimensional signals for camera-based physiological sensing, as well as the capability and limitations of existing camera-based physiological sensing solutions when applied in a driver monitoring system application context.

I. INTRODUCTION

In 2022, land transport accidents were one of the top three leading causes of death in Australia for persons 1-44 years of age [1]. 20%-30% of these accidents are attributed to fatigue [2] and an additional 16% are attributed to distraction [3]. Seeing Machines (SM) is a Tier Two automotive company focused on creating computer vision-based active driver monitoring systems (DMS) with the mission of eliminating transportation fatalities and injuries caused by distraction and fatigue.

Inclusion of physiological signals is a crucial step to the development of more capable DMS. For example; [4], [5], [6] demonstrate there is a 7.5 to 10 beats-per-minute (BPM) difference in heart rate between drowsy and alert states. Furthermore, [7] found a strong correlation between heart rate variability (HRV) and cognitive load, while [8] found that HRV can be used for intoxication detection. Lastly, numerous studies have demonstrated the capacity of camera-based heart rate sensing—specifically, the remote photoplethysmography (rPPG) signal—for spoof detection [9], [10], [11]. Accordingly, inclusion of physiological signals is expected to foster significant improvements and advances in existing DMS technology.

This project seeks to evaluate the capabilities of near-infrared (NIR) coherent structured light emitters for use in

rPPG. While previous research has primarily focused on traditional incoherent illumination sources, the narrow linewidth of coherent sources provides a significant advantage in ability to reject interfering external light. Furthermore, structured light emitters also offer the ability to deterministically derive depth data [12], which could be valuable for detecting respiration rate. Moreover, structured patterns have previously proven to be effective for decoding the ballistocardiogram (BCG) signal [13]. Finally, coherent structured light emitters have been proven to be capable of material detection [14] which could help guide models to the most suitable regions of an image (i.e., skin pixels) for obtaining the rPPG signal.

In particular, a dot projector coherent illumination source delivers all these benefits while still retaining the underlying flood illumination profile found in traditional sources—albeit at a lower intensity. This project successfully extracts and applies these signals to detect the rPPG physiological signal in the context of DMS applications, demonstrating their value for training rPPG deep learning models.

Broadly summarising, the project scope covers (1) the development of a novel system for rPPG data collection; (2) collection of 7 hours of high bit-depth video training data including synchronised physiological signals for truthing; (3) benchmarking this new dataset against existing state-of-the-art rPPG deep learning algorithms [15], [16], [17], [18]; (4) development of novel deep learning algorithms using structured coherent illumination source signals; and (5) validation of these models using the newly compiled dataset. The findings of this project provide valuable insights into the capacity of existing Seeing Machines DMS for rPPG detection, and will guide the future direction of Seeing Machines rPPG research.

II. BACKGROUND

A. Photoplethysmography

As early as 1942 [19], it has been shown that Hb is primarily responsible for the absorption spectra of blood in the visible light region. The absorption spectrum of oxygenated and deoxygenated Hb are significantly different (Fig. 1). It is this difference that forms the basis of photoplethysmography

(PPG). The goal of rPPG, also known as imaging photoplethysmography (iPPG), is to detect this difference via remote sensing means (i.e., a camera), and thus, detect a subjects heart beat.

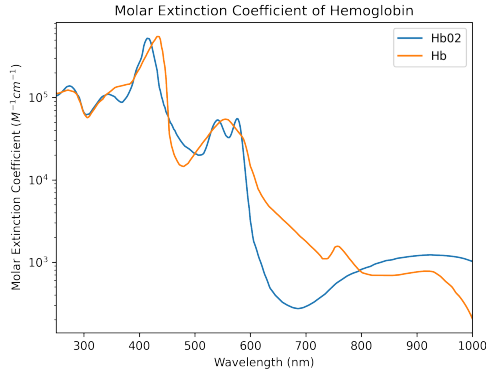


Figure 1. Molar extinction coefficient for oxygenated (HbO_2) and deoxygenated (Hb) hemoglobin [20].

In the NIR region, the absorption of hemoglobin is significantly less intense and suspended blood constituents play a more appreciable role in the absorption spectrum of blood. When compared to visible light, this makes NIR rPPG significantly more difficult, generally requiring more capable hardware (e.g., higher bit-depth). Conversely ambient sunlight rejection is significantly better in the NIR region, resulting in algorithms that are more robust to varying ambient illumination conditions. Furthermore, NIR light is invisible to the human eye making it a valid, non-invasive solution that can be used at night.

B. Camera-based Vital Measurement

Remote camera-based vital sensing seeks to harness subtle variations in the human body to detect a subjects physiological signals.

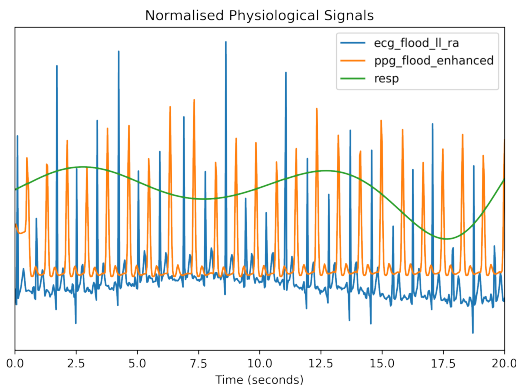


Figure 2. The electrocardiogram (ECG), PPG and respiratory physiological signals

Traditional signal-processing video-based physiological sensing frameworks begin with face landmark detection, followed by segmentation, colour space transform, and filtering

of the desired physiological signal [21], [22], [23], [24]. Such pipelines are difficult to implement, computationally expensive, and rely on handcrafted feature extractors that generalise poorly.

State-of-the-art methods have shifted to deep-learning based approaches [15], [16], [17], [18]. These models are more robust to heterogeneous illumination and other noise factors like subject movement; outperforming complex, handcrafted feature extractors. Furthermore, these deep learning-based approaches have the capacity to be entirely end-to-end solutions, thereby saving significant compute resources and running in real-time on embedded targets.

III. LITERATURE REVIEW

A. Camera-based Vital Measurement

DeepPhys. [15]

In 2016, Chen and McDuff introduced DeepPhys, a convolutional neural network (CNN) architecture for rPPG. DeepPhys is a seminal model in the rPPG research space, having been the first deep-learning approach to significantly outperform unsupervised methods. Chen and McDuff were motivated by the idea of an end-to-end model that reads motion information from video frames and jointly learns a spatial mask to detect regions-of-interest and recover the rPPG and respiration signals. At time of inception, this was a unique idea as existing rPPG algorithms were solely focused on recovering the rPPG signal embedded in temporal variations of pixel intensity [25], [24], [21], [22].

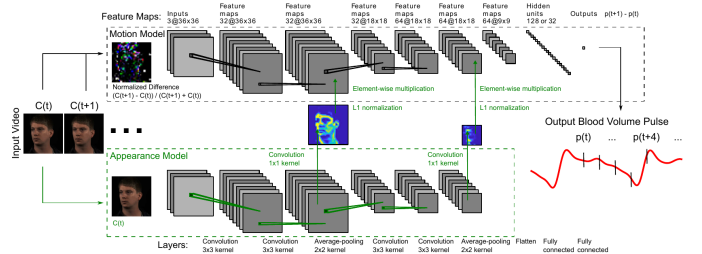


Figure 3. DeepPhys architecture summary [15].

The basic principle of DeepPhys is to use sequential difference frames to generate a motion representation that can better capture physiological motions under heterogeneous illumination. This motion representation is jointly learned alongside an auxiliary appearance representation that acquires attention from the human appearance. Both branches use a VGG-style backbone [26]. Notably, this approach reverses the traditional use of attention, which typically derives attention from motion to guide the learning of appearance representation. The output of the appearance branch is a set of masks that are convolved with the motion branch to assist the motion learning. The attention masks help the network minimize the negative effects introduced by motion and lighting noise.

Accordingly, the inputs to DeepPhys are raw video frames (appearance branch inputs) coupled with normalised difference frames (motion branch inputs). These frames are downsampled

to L pixels by L pixels using bicubic interpolation. The authors found $L = 36$ to be a good trade-off between suppressing camera noise while retaining spatial information.

The training signal for DeepPhys is the derivative of the PPG signal. The objective of the model is to predict this differentiated PPG signal using the input data. As is standard with papers in the physiological sensing space, heart rate is computed by taking the dominant frequency component of the physiological signal. Comparatively, the use of a mean-squared error loss function means the model is trained to minimise the temporal error. Although these generally have high correlation, a small temporal error does not guarantee a small frequency error. To counteract this, the authors use an ensemble learning approach where models were trained for an extra 16 epochs after convergence. These models were applied to the training data and the model with the smallest frequency error was chosen.

On all tasks of the AFRL [27] and MAHNOB-HCI [28] datasets, DeepPhys was able to significantly outperform traditional handcrafted signal-processing approaches like ICA [29], [30], and PBV [31] (an extension of CHROM [21]). For example, on the rapid random head movement task of the AFRL dataset, DeepPhys outperformed ICA and PBV by more than ten times (13.4 and 10.3 versus 1.78 BPM MAE). Furthermore, the authors demonstrate their model is capable of running on monochromatic NIR videos, achieving a MAE of 0.55 BPM on the MAHNOB-HCI dataset.

MTTS-CAN. [16]

MTTS-CAN was the first application of a temporal-modeling to the rPPG space. Temporal shift modules (TSMs) allow CNNs to capture complex temporal dynamics with the computational efficiency of 2D convolutions [32]. This is of particular relevance to rPPG, where high frame-rates (at least 100 Hz) are required for the precise measurement of arterial fibrillation [33], hypertension [34], and heart rate variability [35]. These were the principles that guided Liu, Fromm, Patel, and McDuff in the construction of MTTS-CAN, a novel multi-task temporal shift convolutional attention network that runs at 150 frames-per-second (FPS) while achieving state-of-the-art performance on benchmark datasets.

The optical model proposed by the authors extends upon that which is outlined in [15] by leveraging the interdependence of the respiration and pulse signals to define a new rPPG signal that is a combination of imaging BCG (iBCG), rPPG, and respiration. Since the rPPG and respiration signal are closely intertwined, the authors argue a temporal multi-task learning approach seems optimal as it could leverage the cross-correlation of the two signals.

The architecture of MTTS-CAN is identical to DeepPhys [15], with the addition of temporal-shift modules (TSMs) before each conv2D block in the motion branch. This allows the model to efficiently capture complex temporal dynamics that extend beyond those within a single set of consecutive frames—i.e., difference frames. Unlike DeepPhys, the input to the appearance branch is a single averaged frame. This is done so that the attention mask (1) need only be calculated

once, (2) captures most of the pixels that contain skin, and (3) reduces camera quantisation error.

Through these changes, MTTS-CAN operates at 150 FPS, while achieving state-of-the-art prediction performance. In particular, the authors find that the addition of TSMs leads to greater resilience to noise (e.g., motion, varying ambient illumination, etc.). Through a systematic evaluation on videos with varying velocities of rotational head motion, the addition of TSMs is shown to perform strongly on tasks with greater velocity head motion. Moreover, although tensor shifting provides important temporal information, it also introduces extra noise. Despite this, the authors find that the attention module is effective at separating the desired signal from the added noise.

EfficientPhys. [17]

EfficientPhys is a lightweight architecture for camera-based vitals measurement with an emphasis on low-latency performance. Prior neural rPPG methods rely on hand-crafted pre-processing steps like normalised difference frames [15], [16]; and traditional signal-processing methods rely on facial landmark detection, segmentation, colour space transforms, and pixel averaging. These pre-processing steps are computationally expensive and prevent the network from learning these features in a data-driven manner. Furthermore, most of these steps are non-trivial to implement and optimise for real-time performance on an embedded target but privacy preservation is crucial for camera-based physiological sensing meaning cloud-based processing is strongly undesirable.

EfficientPhys seeks to address these deficiencies by building upon pre-existing research in the rPPG deep learning space (primarily DeepPhys [15] and MTTS-CAN [16]) while seeking to provide a truly end-to-end solution that requires no pre-processing steps and can be practically run in real-time on a mobile device while maintaining state-of-the-art performance on heterogeneous scenes.

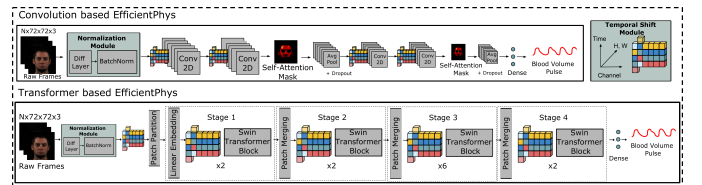


Figure 4. EfficientPhys architecture summary [17].

The authors propose two variants of EfficientPhys: one with a convolutional architecture and another with a transformer architecture. The input to either variant is unprocessed video frames without accurate face cropping. Before the stem layer of each model, a normalisation module is inserted that performs equivalent pre-processing to what is described for DeepPhys [15]. The primary difference between the two is that the batchnorm implementation allows the model to learn the optimal normalisation parameters that can magnify the contained physiological signals in comparison to a bespoke hand-crafted normalisation operation.

In designing the convolutional network architecture, the authors effectively flattened the two branches of MTTSCAN into a single branch end-to-end network. This is done under the assumption that the attention masks can be learned through self-attention modules, rather than requiring a separate appearance branch. They are softmax attention layers with 1D convolutions followed by a sigmoid activation function. Then, normalisation is applied to remove the outlier values in the attention mask, and the final normalised attention mask is element-wise multiplied with the output from the tensor-shifted convolution.

Comparatively, the transformer-based architecture replaces the conv2D blocks with 2D Swin transformer blocks [36]. Inspired by the idea of shifting spatial window partitions, the authors add TSMs before each Swin transformer block to facilitate information exchange across the temporal axis. The TSM modules give the architecture the ability to perform efficient spatial-temporal modeling and attention by combining shifting window partitions spatially and shifting frames temporally. Notably, the TSMs do not introduce any learnable parameters.

The authors found the convolutional-based architecture to significantly outperform the transformer-based architecture in both accuracy and latency. Performance further degraded when the transformer architecture was shrunk to use a similar number of parameters to the convolutional architecture. These results indicate shallow transformer architectures struggle to model subtle changes of skin pixels. However, the authors do acknowledge transformers typically require larger datasets to achieve state-of-the-art results while there is a limited quantity of data available in the rPPG research space.

IV. DATASET

A. SM-EOD

Datasets in the rPPG research field are limited, primarily due to privacy concerns related to their collection and distribution. Commonly used datasets include AFRL [27], UBFC-rPPG [37], PURE [38], UBFC-PHYS [39], MMPD [40], SCAMPS [41], VIPL-HR [42], and MAHNOB-HCI [28]. However, accessing these datasets can be challenging. For instance, AFRL lacks clear instructions on how researchers can obtain access, while datasets like MMPD require a formal request for use. Importantly, all of these datasets prohibit commercial use. Furthermore, none of these datasets include structured light illumination sources. Consequently, it was necessary to create a new dataset consisting of synchronised camera and physiological data.

1) *Configuration*: The dataset was collected from two viewpoints: a narrow-field of view camera positioned at the IC and a wide field-of-view camera located at the RVM position.

Seeing Machines uses a proprietary automatic exposure controller (AEC) to ensure consistent subject appearance under heterogeneous external illumination. In theory, such a controller could significantly degrade rPPG performance as its objective would be to undo the variations in pixel intensity caused by the rPPG signal. After some testing, it was verified that the AEC controller was not sensitive to the rPPG signal

though it was still expected that the AEC controller would degrade rPPG performance. Accordingly, an additional camera was added to the instrument cluster position that did not have an AEC (i.e., was operating with static exposure and gain). This camera provided data most representative of existing public rPPG datasets.

Table I: Summary of the three cameras used in the project's data collection campaign.

	Camera 1	Camera 2	Camera 3
Location	IC	RVM	IC
HFOV (°)	35	160	45
VFOV (°)	25	130	35
Resolution (MP)	5		2.4
Bit depth (bits)	10		12
Frame-rate (fps)	60		
Illumination	4x 940nm LED + 940nm dot-pattern projector		N/A
AEC	✓		✗
3D capable	✓		✗
Bandpass filter (nm)	940 ± 50		
Effective working distance (mm)	750		
MTF50 (lp/mm)	65	77	

For monitoring of the physiological signals, a Shimmer 3 GSR+ and ExG unit were used with sampling rate set to 1024 Hz. They were controlled using the Shimmer LabVIEW API [43] and synchronised with the recorded video data in post-processing.

2) *Protocol*: In collecting this dataset, one of the key objectives was to compile a dataset that would be of significant utility to future physiological sensing endeavours at Seeing Machines. Ideally, the protocol should be comprehensive and consider a variety of behaviours that satisfy the needs of this project while considering behaviours that might be desirable for future projects.

Traditional public rPPG datasets focus on motion and/or heterogeneous external illumination noise factors. These datasets fail to capture the diversity of noise factors encountered in a DMS application context, which include: motion, heterogeneous external illumination, makeup, facial hair, skin pigment, masks, eyewear, headwear, intoxication, medication (e.g., stimulants), pacemakers, and more. Accordingly, a protocol was devised that sought to capture a more comprehensive variety of noise factors. The other objective was to compile a dataset that covered behaviours similar to those in public datasets to allow for meaningful comparison of the projects results.

Typical Subject.

Inspired by the AFRL dataset [27], a variety of tasks with varying degrees of movement were included in the protocol. For a typical subject, three unique guided recordings were specified:

- **iGR1**: Subject fixes their gaze on a target and maintains static head position. Five gaze targets were defined: the instrument cluster, the center console, on-road, the left-side mirror, and the right-side mirror.

- **iGR2**: Subject continuously moves their head position for 2 minutes in a random pattern.
- **iGR3**: Subject repeatedly leans forwards and backwards for 1 minutes.

Lastly, data representative of a typical DMS scene was captured. Namely, a naturalistic recording configuration (**iNR1**) was defined that required the subject to drive for 10 minutes uninterrupted along the Tokyo Shuto Expressway using the simulator rig. Behaviour during this configuration is entirely naturalistic, meaning each video may contain any combination of static, moving, and talking behaviour.

The data collection procedure for a typical subject is shown in Fig. 5.

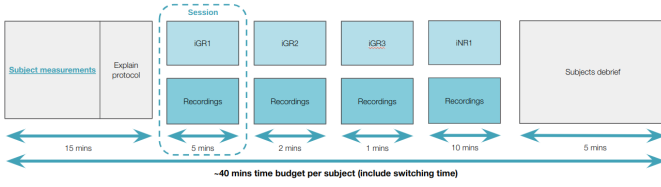


Figure 5. Data collection procedure for a typical subject.

Noise Factors.

Makeup products like foundation and contour have a significant impact on the transmission and reflectance of skin [44]. Existing research has shown the reduced transmissivity of cosmetically-enhanced skin degrades the signal-to-noise ratio of the rPPG signal. Although the impact of makeup on rPPG signal strength is more severe in the visible spectrum, a reduction of 54.5% to 61.4% is still observed in the NIR spectrum [45]. To investigate this impact, a configuration was defined requiring a subject to cover half their face (split vertically down the center) with makeup. The subject was then instructed to stare directly at the instrument cluster for a two minute recording (**iGR4**).

Furthermore, the distribution of average resting heart rate amongst the general population is fairly narrow. As shown in the results of this paper, on the MR-NIRP rPPG public dataset, it is possible to score a MAE of less than 10 BPM by guessing the mean heart rate of the entire dataset. Accordingly, for our dataset, we deemed it important to include a noise factor where a subject is required to elevate their heart rate to 70%-80% of their maximum heart rate (determined using the Karvonen formula) before commencing recording. For exercised subjects, three additional recording configurations are specified which are a repetition of the typical guided behaviour recordings with the subjects heart rate elevated to zone three.

Lastly, existing DMS systems have demonstrated significant efficacy in detecting fatigue and reducing the number of overall fatigue-induced accidents [46]. The next technological revolution in the DMS space will be the capability to reliably detect and prevent distraction and intoxication related accidents. Existing research has sought to use heart rate statistics like HRV to infer cognitive load [7] and intoxication

[8]. Accordingly, an additional final naturalistic configuration, **iNR3**, was defined in which subjects were required to reach a blood alcohol concentration >0.08 (verified using a calibrated breathalyser) before performing 10-minutes of naturalistic driving.

Importantly, a baseline set of guided recordings was taken for all noise subjects.

3) *Post-processing*: Following conversion to EOD, the data was uploaded to an Amazon s3 bucket. At this point, the total dataset size exceeded 40 TB, meaning extraction of the desired training features posed a significant logistical issue. A windowing approach was used where an EOD was read-in, processed and saved in 10-second intervals. This also provided an opportunity for judgement of the recorded ground-truth PPG signal quality. To judge the signal quality, the HeartPy Python package [47] was used to and derive an enhanced version of the raw PPG signal and compute the RMSSD¹ of each window. If the RMSSD was greater than 130 ms, the window was shifted 1-second forward. This process was repeated until a good window was found, at which point the data was processed and saved, and the window was then shifted by a full 10-seconds. This Python script was deployed with AWS Batch Jobs, allowing the entire dataset to be processed in parallel. This meant the total time to process and extract the desired training features from the dataset **was reduced to just 3 hours**. A visualisation of the extracted training signals for a 3D camera (i.e., the two 5 MP cameras) is shown in Fig. 6.

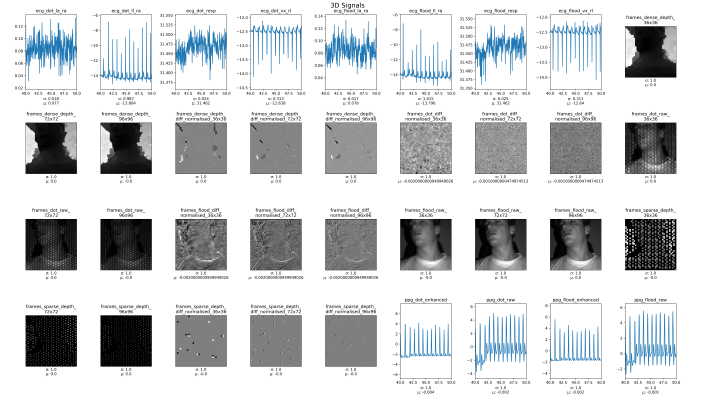


Figure 6. The extracted 3D training features for the IC 5MP camera.

B. MR-NIRP Car Dataset

The MERL-Rice Near-Infrared Pulse (MR-NIRP) Car Dataset [49], is designed for camera-based vital signs estimation using NIR imaging during driving. The dataset includes face video recordings collected in narrow-band 940 nm NIR, with pulse oximeter data providing ground-truth vital signs.

¹The root-mean square of successive differences between consecutive RR intervals. The typical range of RMSSD values is 25 ms to 80 ms [48]. A RMSSD significantly higher than this provides a good indication the PPG signal is corrupted with noise artefacts.

It contains data from 18 subjects, including both male and female participants with diverse skin tones, recorded in various driving conditions to analyse the impact of motion and ambient light on rPPG in a practical scenario. Four recordings were available per subject, corresponding to conditions “still” and “small motion”, with environments “garage” and “driving”. Considering only these recordings, the total dataset duration was three hours.

For this project, the MR-NIRP dataset was preprocessed using the AWS Batches framework described above and used for both independent training and cross-dataset validation. Note that the dark frames included in the MR-NIRP dataset were not used for training.

V. METHOD

Each model was trained on the SM-EOD and MR-NIRP datasets. Cross-dataset validation was performed, where models were either trained and tested on the same dataset or trained on one dataset and tested on the other. Importantly, to our knowledge, this paper marks the first time these neural methods have been applied to the MR-NIRP dataset.

For all experiments, models were trained for 30 epochs. In each case, the epoch with the best validation score was used for testing. An ablation analysis was performed where models were trained with Negative Pearson [50], mean-squared error (MSE), and L1 loss. The exception is PhysFormer models, which were trained using the PhysFormer loss function.

On the SM-EOD dataset, models were trained with learning rates (LRs) $9e-4$ and $9e-5$. On the MR-NIRP dataset, models were trained with LRs $9e-3$ and $9e-4$. All models are trained using a 1cycle LR policy [51], with the exception of the PhysFormer models which did not use a LR policy. All models were trained using an AdamW optimiser [52]. For all models, a five-fold analysis is performed. In the case that a model is trained and tested on the same dataset, the average performance across the five-folds is recorded. When trained on one dataset and tested on another, the model with the best performance was selected, as the folds were applied only to the train and validation data, while the test dataset remained intact.

All models were trained using inputs of resolution 36x36, with the exception of PhysFormer which was trained using 96x96 resolution inputs that were zero-padded to 128x128 to match the input size expected by the architecture. Although high bit-depth data was recorded, the results of this project were generated using solely 8-bit input data. For all training and test runs, the makeup **iGR4** and exercised **iGR5** configurations were excluded unless otherwise stated. Unless otherwise stated, all results from the SM-EOD dataset were generated using the 2.4 MP IC camera.

A. Metrics

The ground-truth labels and model predictions were detrended and filtered using a first-order Butterworth bandpass filter with cutoffs at [0.75, 2.5] Hz (corresponding to [45, 150] BPM). Heart rate was measured in the frequency domain,

with 10-second buffers of prediction/label data processed for each heart rate measurement. The heart rate for each filtered window was determined by the frequency with the peak spectral power. For each method, the mean absolute error (MAE) and mean absolute percentage error (MAPE) between predictions and ground truth for 10-second non-overlapping windows was evaluated.

VI. RESULTS AND DISCUSSION

A. SM-EOD and MR-NIRP

All Behaviours.

Table II: Results for models trained and tested on the corrupted SM-EOD dataset and the MR-NIRP dataset. The corrupted dataset cells are shaded in red.

Method	Train Set	Test Set			
		SM-EOD	MR-NIRP [49]	MAE↓	MAPE↓
LGI [24]	N/A	25.05	22.27	14.56	22.87
Guess Mean HR	N/A	8.85	12.30	9.19	13.27
DeepPhys [15]	SM-EOD	13.99	17.17	10.77	14.94
	MR-NIRP [49]	7.11	9.00	3.75	4.70
TS-CAN [16]	SM-EOD	14.37	18.07	12.01	16.91
	MR-NIRP [49]	5.62	7.01	3.92	4.83
EfficientPhys [17]	SM-EOD	14.60	17.98	11.55	16.28
	MR-NIRP [49]	8.11	10.34	3.56	4.91
PhysFormer [18]	SM-EOD	30.62	44.31	8.95	13.55
	MR-NIRP [49]	10.60	13.64	5.95	7.10
DeepPhys-ConvNeXt	SM-EOD	13.28	16.46	8.31	10.63
	MR-NIRP [49]	8.31	10.63	8.40	10.47
DeepPhysFormer	SM-EOD	12.69	15.91	13.25	18.34
	MR-NIRP [49]	6.50	7.96	4.72	5.73
DeepPhys-Dot	SM-EOD	14.98	19.54		
TSCAN-Dot	SM-EOD	15.10	19.68		
EfficientPhys-Dot	SM-EOD	15.88	20.73		
PhysFormer-Dot	SM-EOD	37.82	54.39		
DeepPhysFormer-Dot	SM-EOD	17.18	21.57		

Results on the SM-EOD dataset were consistently poor. After some investigation, it was uncovered that the software developed to record the synchronised video and physiological signal data suffered a bug that meant **none of the ground-truth data was reliably synchronised**. In short, the capture software assumed that each sample of the physiological data was exactly $T = \frac{1}{1024}$ seconds apart, but the physiological sensors transmitted over a Bluetooth channel meaning packet loss was inevitable. A single dropped packet was enough to compromise the synchronisation of the recording and without knowledge of when packets were dropped, it was not possible to properly synchronise the ground-truth data. This finding was very disappointing as it compromised the results of this project, but it does explain the discrepancy in performance observed between the SM-EOD and MR-NIRP [49] datasets. Results for models trained and tested on the MR-NIRP dataset and the corrupted SM-EOD dataset are shown in Table II.

Guessing the mean heart rate of either dataset yields better performance than any of the models trained on SM-EOD dataset. Comparatively, performance on the MR-NIRP dataset is consistently good, with errors as low as 3.56 BPM MAE when considering all behavioural and environmental settings. Even when testing on the corrupted ground-truth SM-EOD

data, models trained on the MR-NIRP dataset still achieve respectable results. This is because the time-varying phase shift of the SM-EOD ground-truth signal compromises the utility of the dataset for training, but does not severely compromise validation capacity—particularly the beginning of each recording before any substantial packet loss is experienced. An example of the predictions for the TS-CAN model trained on the MR-NIRP dataset and tested on the corrupted SM-EOD dataset are shown in Fig. 7. Even though the instantaneous magnitude of the predicted rPPG and corrupted PPG labels are weakly correlated, the frequency content of the two signals is similar (i.e., they have the same number of peaks for a given window). Accordingly, the corrupted data can still be used for validation.

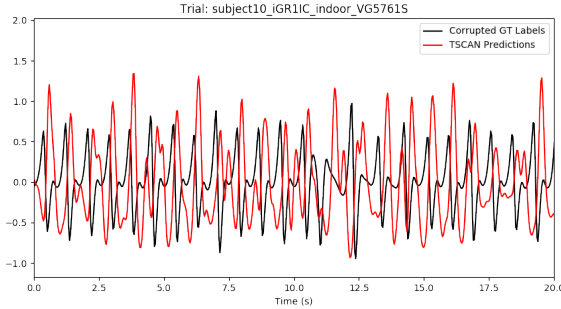


Figure 7. Plot of the TSCAN predictions and corresponding corrupted ground-truth signal for an **iGR1** static behaviour recording. A time-varying phase shift is observed in the corrupted ground-truth data due to unaccounted packet loss.

Behaviour Specific Results.

Due to the corrupted ground-truth signals in the SM-EOD dataset, analysis in this section only examines the MR-NIRP dataset. In Table III, behaviour specific results are shown for models trained and tested on the MR-NIRP dataset. Best results are obtained for the indoor static behavioural configuration. This is not unexpected, as prior research has shown that motion and heterogeneous illumination are the two biggest noise factors for rPPG sensing [22], [23], [21], [24], [15].

Most models exhibit similar performance for the garage small motion and driving still recording configurations, with a small degradation in performance for the driving still configuration. However, [23] finds that the average movement of each subject in the MR-NIRP dataset is slightly larger in the driving still configuration versus the garage small motion configuration due to the influence of vehicle momentum on the passenger kinematics. Accordingly, these results seem to indicate that heterogeneous illumination is a significantly less impactful noise factor than motion for outdoor rPPG performance when using a narrow bandpass filter around 940 nm as employed within the MR-NIRP optical pathway. One caveat to this is that no measurements were taken of ambient illumination intensity for the MR-NIRP dataset, so it is not possible to quantitatively validate this conclusion.

A significant degradation in performance is observed for all models for the driving small motion configuration. This result suggests additional training data or architectural changes will be necessary for neural methods to robustly operate in a DMS rPPG application context. For example, incorporation of temporal filtering of model outputs could harness the property that heart rate is typically a wide sense stationary signal to produce more robust predictions in the face of motion. An observer (e.g., Kalman filter) seems like the most obvious solution here. Although this would likely significantly boost performance under normal circumstances, such approaches would introduce complexity in ensuring reliable prediction in the face of legitimate rapid increases in heart rate (e.g., myocardial infarction).

DeepPhys performs strongest in the simplest behavioural condition, but significantly degrades with the addition of motion. TS-CAN shows strong performance in the presence of motion (particularly the driving still configuration). This result is supported by [16], who found the addition of TSMs significantly improved performance on tasks with greater velocity head motion. EfficientPhys performs poorly with the addition of motion, however, demonstrates robustness to heterogeneous illumination. This results suggests the inclusion of the difference and batch norm layers in the model stem allows it to learn a normalisation more robust to heterogeneous illumination than the handcrafted normalisation employed by DeepPhys and TS-CAN.

The PhysFormer model consistently underperforms all models. This can be attributed to a lack of sufficient training data (PhysFormer is a large transformer architecture meaning it is expected to require more training data than the other CNN-based architectures). This analysis is supported by [17], who found transformer architectures were consistently outperformed by CNN equivalents for the same reasons. Another consideration is the ablation analysis used for the training regimes. The best performance for all CNN models was obtained from LR $9e-3$, while PhysFormer was consistently better with LR $9e-4$. Furthermore, the training curves for the PhysFormer models did not converge until epochs 25 to 28, while the CNN models all converged before epoch 20. It is expected that training PhysFormer with an even smaller LR and for more epochs could improve performance.

The DeepPhys-ConvNeXt model demonstrates equivalent performance to the DeepPhys model, with a small boost in performance for the garage small motion configurations. Notably, however, the use of ConvNeXt blocks means this model has significantly more parameters than the DeepPhys model (102.797 MFLOPs versus 21.7791 GFLOPs, respectively), resulting in a much slower inference time. Finally, the DeepPhysFormer model is consistently outperformed by the DeepPhys model, suggesting that the PhysFormer loss function [18] translates poorly to the DeepPhys architecture.

B. Pseudo-Labelled SM-EOD and MR-NIRP

Due to the corrupted ground-truth data, it was not possible to infer any meaningful results by training on the raw SM-

Table III: Behaviour-specific results for models trained and tested on the MR-NIRP dataset.

Method	Train Set	Test Set							
		MR-NIRP [49]							
		Behaviour							
		garage still		garage small motion		driving still		driving small motion	
		MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓
DeepPhys [15]	MR-NIRP [49]	0.09	0.10	1.94	2.63	2.01	2.73	8.33	10.13
TS-CAN [16]	MR-NIRP [49]	0.35	0.47	1.49	1.89	0.36	0.43	7.63	9.36
EfficientPhys [17]	MR-NIRP [49]	0.32	0.47	4.35	5.96	1.61	2.27	6.64	8.91
PhysFormer [18]	MR-NIRP [49]	2.55	2.60	10.37	11.32	4.56	5.64	7.27	8.95
DeepPhys-ConvNeXt	MR-NIRP [49]	0.09	0.10	1.49	1.91	1.48	1.92	10.17	12.65
DeepPhysFormer	MR-NIRP [49]	0.40	0.54	1.93	2.63	3.19	3.98	10.04	11.97

EOD dataset. However, as demonstrated in Table II, most models were able to achieve respectable performance on the MR-NIRP dataset. In particular, as shown in Table III, most results on the static indoor condition of the MR-NIRP dataset are less than 0.5 BPM MAE. These results suggest that models trained on the MR-NIRP dataset could be used to generate pseudo-labels for the SM-EOD dataset—at the very least for static behavioural configurations. These results are the topic of this discussion.

The TSCAN model trained and tested on the entire MR-NIRP dataset achieving a MAE of 5.62 BPM across all conditions of the corrupted SM-EOD dataset was used to generate pseudo ground-truth PPG labels for the SM-EOD dataset. These predicted labels were detrended and bandpassed filtered before use. An example of the generated pseudo-labels and corrupted underlying ground-truth signal is shown in Fig. 7. Because the performance of this model was significantly worse on moving configurations in the MR-NIRP dataset (Table III), it was not appropriate to use such recordings. Accordingly, models were only trained on the static behaviours (**iGR1** recordings) of the SM-EOD dataset. These models were then tested on the static behaviours of the pseudo-labelled SM-EOD dataset and the entirety of the MR-NIRP dataset. Results are shown in Table IV. No results were generated for models trained on the MR-NIRP dataset and tested on the pseudo-labelled SM-EOD dataset as they would not provide any valuable insights.

All Behaviours.

In the case of the SM-EOD dataset, at best, these results indicate that models as good as the TSCAN model used to generate the pseudo-labels could be trained using the SM-EOD dataset with properly synchronised ground-truth. These results are not particularly interesting, with the exception of the results recorded for 3D signals which provide insight into the viability of dot and depth frames for rPPG. Of more interest is the results obtained by training on the pseudo-labelled SM-EOD dataset and tested on the MR-NIRP dataset which has proper ground-truth. These results provide insight into the capability of the developed data collection system and optical pathway for sensing of the rPPG signal assuming the ground-truth signal is properly synchronised.

Considering results obtained by training and testing on

Table IV: Results for models trained and tested on the pseudo-labelled SM-EOD dataset and the MR-NIRP dataset. The SM-EOD test and training sets consist of only the **iGR1** behavioural recordings.

Method	Train Set	Test Set			
		SM-EOD (iGR1 only)		MR-NIRP [49]	
		MAE↓	MAPE↓	MAE↓	MAPE↓
LGI [24]	N/A	15.24	18.73	14.56	22.87
Guess Mean HR	N/A	9.10	13.04	9.19	13.27
DeepPhys [15]	SM-EOD (iGR1 only)	0.64	0.97	7.38	9.88
	MR-NIRP [49]			3.75	4.70
TS-CAN [16]	SM-EOD (iGR1 only)	0.31	0.48	5.23	7.38
	MR-NIRP [49]			3.92	4.83
EfficientPhys [17]	SM-EOD (iGR1 only)	0.85	1.27	6.93	9.71
	MR-NIRP [49]			3.56	4.91
PhysFormer [18]	SM-EOD (iGR1 only)	7.64	10.51	5.18	8.57
	MR-NIRP [49]			5.95	7.10
DeepPhys-ConvNeXt	SM-EOD (iGR1 only)	0.66	0.99	9.53	12.75
	MR-NIRP [49]			4.25	5.31
DeepPhysFormer	SM-EOD (iGR1 only)	1.01	1.47	8.40	11.46
	MR-NIRP [49]			4.72	5.73
DeepPhys-Dot	SM-EOD (iGR1 only)	0.90	1.34		
TSCAN-Dot	SM-EOD (iGR1 only)	0.94	1.40		
EfficientPhys-Dot	SM-EOD (iGR1 only)	1.44	2.03		
PhysFormer-Dot	SM-EOD (iGR1 only)				
DeepPhysFormer-Dot	SM-EOD (iGR1 only)	1.16	1.71		
DeepPhys-FloodDotDepth	SM-EOD (iGR1 only)	0.95	1.40		
DeepPhysFormer-FloodDotDepth	SM-EOD (iGR1 only)	2.23	2.86		

the pseudo-labelled dataset, the TS-CAN model was most effective at reproducing the pseudo-label signal. This result is not surprising considering the pseudo-labels were generated using a TS-CAN architecture model. In particular, the TS-CAN and PhysFormer models trained on the pseudo-labelled SM-EOD dataset are within 1 BPM MAE of their counterparts trained on the MR-NIRP dataset, with the PhysFormer model actually surpassing its counterpart. These results suggest that the use of pseudo-labels was a valid training strategy that has empowered these models to learn to detect the rPPG signal. All models are able to substantially surpass the performance of guessing the MR-NIRP dataset’s mean heart rate.

The *Dot* and *FloodDotDepth* variants both demonstrate a capacity to learn the pseudo-labelled rPPG signal. It was not possible to test these models on the MR-NIRP dataset as there is no equivalent optical pathway in the MR-NIRP dataset. Despite this, the results do indicate that it is possible to isolate the rPPG signal from *Dot* frames using neural methods. No increases in performance were observed for the *FloodDotDepth* variants, contradicting the hypothesis of this project. Despite this, it is important to note that the addition

of depth as an rPPG signal was expected to provide the most benefit in scenarios with movement, but it was not possible to examine such scenarios given that the pseudo PPG signal cannot be considered as a reliable ground truth when the subject is moving. Notably, results for the *Dot* variants were generated using the 5 MP IC camera. This was done as a significant degradation in performance was observed for the 2.4 MP IC camera. This degradation is discussed in Section VI-B1. Results for the *FloodDotDepth* variants were generated using the 5 MP IC camera as this was the only IC camera with depth capabilities.

Behaviour Specific.

In Table V, behaviour specific results are shown for models trained on the static recordings of the pseudo-labelled SM-EOD dataset and tested on the MR-NIRP dataset. A clear degradation is observed under the addition of movement—a similar but more severe degradation as is observed in Table III. These results are supported by [15], who also showed that models trained on only static recordings generalise poorly to recordings containing motion. Similar to Table III, these results indicate that motion is a more significant noise factor for rPPG than heterogeneous illumination. Interestingly, these results suggest that models trained with controlled illumination data generalise well to heterogeneous illumination scenarios. These results support the mathematics shown in [15], which posit that difference normalised frames are an effective method for removing the influence of controlled illumination as well as slow fading external illumination. Despite this, it is important to note that difference normalised frames are not effective for removing fast fading external illumination noise factors (i.e., external illumination varying faster than the Nyquist rate of the difference frame frequency). As aforementioned, the MR-NIRP dataset does not contain measurements of external illumination power for each recording. However, these results, as well as those in Table III suggest that the MR-NIRP dataset is unlikely to contain large quantities of driving data with fast fading external illumination. It is reasonable to assume the performance of all neural methods would degrade significantly for such data.

1) *Higher Resolution Results:* Using dot frames, the 2.4 MP IC camera at 36x36 resolution produced significantly worse results than the 5 MP IC camera at the same resolution on the pseudo-labelled SM-EOD dataset (e.g., 6.31 versus 0.64 BPM MAE for the DeepPhys model). This result raises questions regarding the difference between the two configurations. It was suspected that higher-resolution dot frames would yield better results because the high-frequency content of the dot frames creates artifacts in low-resolution images. This affects the 2.4 MP camera more due to its wider field-of-view, which more severely compresses the dot beam profiles when downsampled to 36x36 pixels as shown in Fig. 8.

To test this hypothesis, models were trained on 96x96 pixel frames from each camera, with results shown in Table VI. Notably, the discrepancy persisted, thereby invalidating the hypothesis. In reality, this discrepancy was actually found to

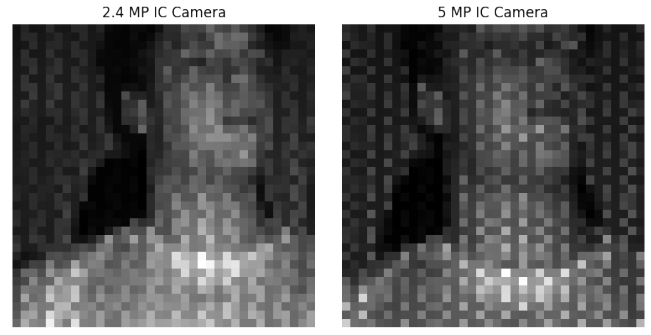


Figure 8. Comparison of the downsampled dot frames for the 2.4 MP and 5 MP IC cameras. More high-frequency artefacts resultant of the downsampling are observed in the 2.4 MP IC camera due to its wider field-of-view.

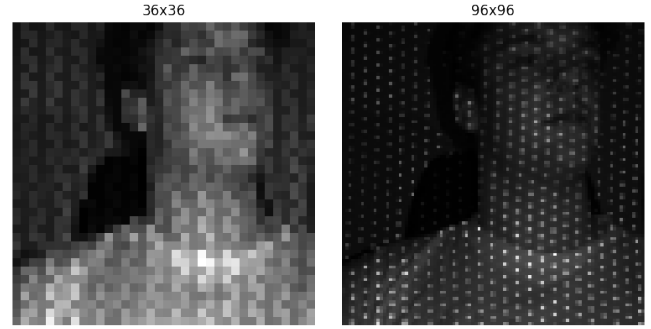


Figure 9. Comparison of the downsampled dot frames for resolutions 36x36 and 96x96 pixels. At 36x36, the beam profiles of individual dots are no longer discernible.

be due to improper synchronisation of the triggered cameras exposure resulting in flickering of the illumination signal.

Furthermore, the performance of most models actually improves for the lower resolution dot frames. These results suggest that the underlying flood profile of the dot pattern projector is a more important signal for sensing the rPPG signal than the actual beam profiles of the dots. However, if we visualise the attention masks of the DeepPhys model for high-resolution inputs (Fig. 10), we can see that the opposite is actually true—the model has learned an attention representation that actively focuses on the dot pixels! Thus, these results suggest that it is possible to isolate the rPPG signal solely based of the signals encoded in a dot pattern projector’s beam profiles (i.e., intensity, dilation, centroid, etc.).

2) *Rear-view Mirror Results:* The RVM poses a significant challenge for rPPG signal monitoring due to several factors: (1) the subject occupies a much smaller portion of the frame, (2) only half of the subject’s face is typically visible during normal driving conditions, and (3) the wide field-of-view lens distorts the subject’s appearance when the RVM is positioned naturally. A comparison between a typical scene for an IC recording versus an RVM recording is shown in Fig. 11. In this section, results are generated for the RVM camera position to

Table V: Behaviour-specific results for models tested on the MR-NIRP dataset.

		Test Set							
		MR-NIRP [49]							
		Behaviour							
Method	Train Set	garage still		garage small motion		driving still		driving small motion	
		MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓
DeepPhys [15]	SM-EOD (iGR1 only)	2.69	3.54	5.74	7.50	5.85	9.20	9.79	12.08
	MR-NIRP [49]	0.09	0.10	1.94	2.63	2.01	2.73	8.33	10.13
TS-CAN [16]	SM-EOD (iGR1 only)	2.64	3.56	4.39	6.28	2.01	2.76	8.64	11.55
	MR-NIRP [49]	0.35	0.47	1.49	1.89	0.36	0.43	7.63	9.36
EfficientPhys [17]	SM-EOD (iGR1 only)	4.20	6.71	3.98	6.07	5.18	7.19	11.74	15.16
	MR-NIRP [49]	0.32	0.47	4.35	5.96	1.61	2.27	6.64	8.91
PhysFormer [18]	SM-EOD (iGR1 only)	0.48	0.63	10.06	15.88	3.37	5.90	6.20	9.83
	MR-NIRP [49]	2.55	2.60	10.37	11.32	4.56	5.64	7.27	8.95
DeepPhys-ConvNeXt	SM-EOD (iGR1 only)	3.37	4.58	5.48	7.67	8.01	11.71	10.96	14.19
	MR-NIRP [49]	0.09	0.10	1.49	1.91	1.48	1.92	8.33	10.13
DeepPhysFormer	SM-EOD (iGR1 only)	3.76	5.12	7.70	10.58	7.01	10.92	11.50	13.92
	MR-NIRP [49]	0.40	0.54	1.93	2.63	3.19	3.98	10.04	11.97

Table VI: Results for the dot models trained and tested on the pseudo-labelled static behavioural conditions of the SM-EOD dataset for varying input image resolution.

		Test Set							
		SM-EOD (iGR1 only)							
		Camera							
		2.4 MP IC				5 MP IC			
		Resolution							
		36x36		96x96		36x36		96x96	
Method	Train Set	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓
DeepPhys-Dot	SM-EOD (iGR1 only)	6.31	9.25	7.39	11.33	0.90	1.34	0.79	1.18
TSCAN-Dot	SM-EOD (iGR1 only)	2.22	2.86	1.21	2.00	0.94	1.40	1.25	1.71
EfficientPhys-Dot	SM-EOD (iGR1 only)	4.05	4.95	3.36	4.27	1.44	2.03	1.61	2.14
PhysFormer-Dot	SM-EOD (iGR1 only)	N/A		12.66	19.08	N/A		9.72	14.55

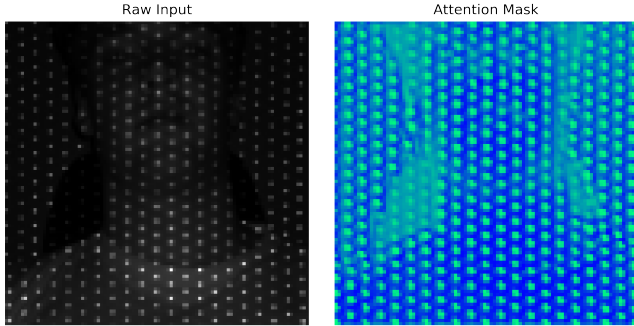


Figure 10. Attention masks produced by the DeepPhys-Dot model for high resolution inputs. The attention mask demonstrates that the model has learned to isolate the rPPG signal using signals encoded in the beam profiles of the dot pattern projector.



Figure 11. Comparison of a 96x96 pixel flood image for an IC recording versus an RVM recording. The RVM configuration poses a significantly more challenging scene for reliable rPPG measurement.

quantify the impact of these challenges on rPPG performance. Results obtained training on the RVM camera configuration with pseudo-labelled ground truth PPG are shown in Table VII.

A small degradation in performance is observed for results obtained from the RVM position. Notably, the AEC was running on the RVM camera meaning that the observed

degradation may even just be attributed to the AEC's impact on the intensity of the illumination source. These results were surprising considering the small portion of pixels occupied by the subject for the RVM configuration. It was suspected that the RVM configuration would benefit from higher resolution inputs, however, the results obtained in Table VII show this to be false. These results provide insight into the importance

Table VII: Results for models trained and tested on the pseudo-labelled static behavioural conditions of the SM-EOD dataset for the 5MP RVM camera.

Method	Train Set	Test Set			
		SM-EOD (iGR1 only)			
		Input Resolution			
		36x36		96x96	
		MAE↓	MAPE↓	MAE↓	MAPE↓
DeepPhys [15]	SM-EOD (iGR1 only)	0.69	1.03	0.77	1.13
TSCAN [16]	SM-EOD (iGR1 only)	0.54	0.81	0.89	1.36
EfficientPhys [17]	SM-EOD (iGR1 only)	0.75	1.09	1.19	1.85
PhysFormer [18]	SM-EOD (iGR1 only)	N/A		10.21	16.53
DeepPhysFormer	SM-EOD (iGR1 only)	0.79	1.14	1.24	1.92
DeepPhys-Dot	SM-EOD (iGR1 only)	0.78	1.16	0.86	1.30
TSCAN-Dot	SM-EOD (iGR1 only)	0.64	0.95	1.05	1.57
EfficientPhys-Dot	SM-EOD (iGR1 only)	0.86	1.27	0.88	1.32
PhysFormer-Dot	SM-EOD (iGR1 only)	N/A		12.22	18.04
DeepPhysFormer-Dot	SM-EOD (iGR1 only)	0.79	1.15	1.24	1.92
DeepPhys-FloodDotDepth	SM-EOD (iGR1 only)	0.69	1.05	0.78	1.17
DeepPhysFormer-FloodDotDepth	SM-EOD (iGR1 only)	2.23	2.86	3.01	3.78

of spatial downsampling for removing image sensor noise, and suggest that further spatial downsampling may actually improve performance for the other camera configurations. This analysis is supported by [53], who are able to successfully isolate the rPPG signal after spatial downsampling down to only 9x9 pixels. Furthermore, these results suggest that rPPG may be possible in OMS systems.

3) *Exercised Subject*.: Most public rPPG datasets fail to capture recordings with a wide range of heart rates. In a DMS application context, this is particularly limiting as one of the desired use cases would be the detection of abnormal events such as cardiac arrest and myocardial infarction. In this section, results are presented for the exercised condition of the SM-EOD dataset for models trained on the pseudo-labelled static conditions and tested using the corrupted ground-truth signal for labelling. In generating the post-processed waveforms for both the predicted and corrupted ground-truth signals, the cutoff frequency of the bandpass filter was raised to [0.75, 3.5] Hz. Results are shown in Table VIII.

Table VIII: Behaviour-specific results for the exercised recording configuration. Models are trained on the static behaviours of the pseudo-labelled SM-EOD dataset and tested on the exercised behaviours of the SM-EOD dataset with corrupted ground-truth signals.

Method	Train Set	Test Set					
		SM-EOD					
		Behaviour					
		iGR5-1		iGR5-2		iGR5-3	
		MAE↓	MAPE↓	MAE↓	MAPE↓	MAE↓	MAPE↓
DeepPhys [15]	SM-EOD (iGR1 only)	68.03	55.48	47.07	44.91	34.80	32.59
TS-CAN [16]	SM-EOD (iGR1 only)	70.40	57.42	41.13	39.25	35.60	33.33
EfficientPhys [17]	SM-EOD (iGR1 only)	75.94	61.94	55.77	53.21	52.21	48.89
PhysFormer [18]	SM-EOD (iGR1 only)	73.56	60.00	28.48	27.17	35.60	33.33

The ground-truth heart rate for the exercised configurations ranges from 100 to 120 BPM, but predictions of all models are consistently in the 60 to 80 BPM range. These results indicate that all models generalise poorly to scenes with higher frequency heart rates than are present in their training data. For all models, better performance is observed for the moving configurations, however, considering the prediction waveforms

(shown in Fig. 12), it is clear that the noisier outputs associated with moving subjects push the peak power spectral density of the model predictions higher. As such, they do not actually reflect better prediction performance. These results suggest that a comprehensive set of training data with a wide range of heart rate values is required for neural rPPG methods to perform robustly in the face of abnormal heart rate events.

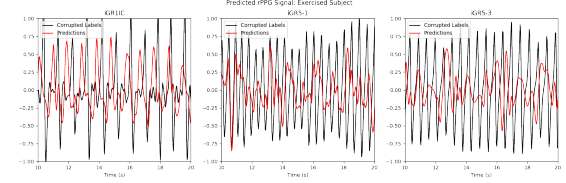


Figure 12. Plot of the DeepPhys rPPG predictions for the exercised recording.

4) *Makeup Impact*: The rPPG signal is around ten times weaker in the NIR spectrum than in visible light [22]. Prior research has demonstrated that the rPPG signal is 54.5% to 61.4% weaker in the presence of makeup [45]. Accordingly, makeup may pose a significant barrier to the application of rPPG in a DMS context.

In Fig. 13, the attention masks of the DeepPhys model trained on the pseudo-labelled SM-EOD dataset are shown for the makeup subject, both with and without makeup. Note that, in the makeup case, only half of the subjects face is covered in makeup. Surprisingly, there is no discernible difference between the attention masks, suggesting that the appearance branch does not learn a representation that prioritises unoccluded skin pixels, but instead learns to isolate the broader face within the scene.



Figure 13. Comparison of an attention mask generated by the DeepPhys appearance branch for a makeup recording versus a typical recording.

In Fig. 14, the predictions of the same model are shown for the makeup subject, both with and without makeup. For both recordings, the subject is static with gaze oriented at the IC camera. Note that the ground-truth signal is the corrupted PPG signal recorded using the finger clip sensor. Only the beginning of the recording is considered, as packet loss is minimal meaning the correlation the prediction and ground-truth labels remains high. Surprisingly, there is not discernible difference in magnitude between the rPPG signal of the subject with and without makeup. This result may be explained by

the fact that only half of the subjects face was covered in makeup. Despite this, these results suggest that neural methods are capable of making rPPG predictions in the presence of makeup, though a more comprehensive test dataset is needed to quantitatively validate this assertion.

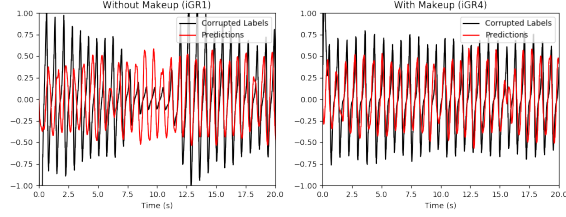


Figure 14. Plot of the DeepPhys model rPPG predictions for the makeup recording.

5) *Spoof Performance*: Prior research has demonstrated that the rPPG signal can be used for spoofing detection [9], [10], [11]. Spoof is a desired feature in DMS—primarily for DMS in the fleet, aviation, and mining industries where unauthorised operators present a significant safety hazard.

During the data collection for this project, a single 10-minute recording of a mannequin head was captured. In this section, predictions from the DeepPhys model, trained on the pseudo-labelled SM-EOD dataset, are evaluated to determine if existing neural rPPG methods can be directly applied to the spoof detection problem. A comparison between the model’s predictions and the ground-truth PPG signal (or lack thereof) is shown in Fig. 15.

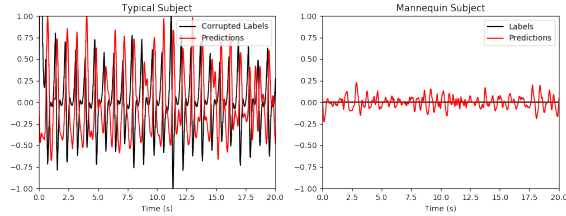


Figure 15. Plot of the DeepPhys rPPG predictions for the spoof recording.

Notably, the model’s predictions are non-zero throughout the spoof recording but are capped around 0.2, about one third of the typical magnitude seen for a human subject. Despite this, it cannot be said that a binary threshold of the rPPG predictions would serve as an effective spoof detection method as the mannequin head is perfectly static for the duration of the recording. It is likely that the magnitude of predictions would change for a moving spoof target (e.g., a person wearing a mask). When considering the predicted waveform for the mannequin head, they bear little resemblance to an rPPG waveform. These results suggest that it may be possible to train a classifier network for spoof detection using the predicted rPPG signal as an input. However, a more comprehensive spoof dataset is required to properly validate this hypothesis.

Notably, the attention mask for the spoof recording bears no obvious differences to those obtained for a regular recording (Fig. 16). These results suggest that the attention branch of the DeepPhys model learns features that isolate the structure of the human face, rather than isolating actual skin pixels. This conclusion is in support of the analysis presented in Section VI-B4.

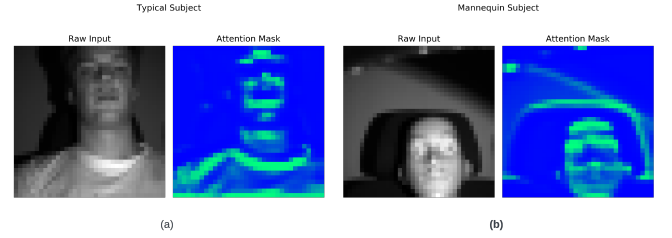


Figure 16. Comparison of an attention mask generated by the DeepPhys appearance branch for a spoof recording versus a typical recording.

VII. CONCLUSION

This project laid the groundwork for rPPG research at Seeing Machines, developing both the infrastructure and methodologies necessary to advance this research topic towards a consumer-grade feature. A comprehensive data collection system was built to capture synchronised physiological and video data. Despite initial setbacks due to corrupted data, this issue has since been rectified and efforts are already underway to extend the protocol for synchronised outdoor data collection, incorporating ambient illumination criteria to enhance the robustness of the dataset. Furthermore, it was shown that although this corrupted data is of no utility for model training it may still be useful for validation purposes.

In addition to data collection, a streamlined pipeline was developed for the efficient extraction of heart rate training features from EODs using AWS Batches. A significant aspect of this project was the adaptation and extension of the rPPG-Toolbox repository, which was modified to support not only the new training feature extraction process but also offer enhanced functionalities such as ablation analysis, attention mask visualisation, metric compilation, neural pseudo-label generation, principal component analysis, ensemble learning, and more.

Through the MR-NIRP dataset, these preprocessing and training pipelines have been validated, confirming the effectiveness of the developed methods. Furthermore, testing on the MR-NIRP dataset using pseudo-labelled data verified the capability of Seeing Machines’ camera systems to perform rPPG measurements.

Although the primary project objective of assessing the viability of three-dimensional signals for heart rate sensing could not be fully realised due to the corrupted ground-truth labels of the training data, preliminary analysis demonstrated that three-dimensional signals indeed offer a valid basis for

rPPG signal extraction. In particular, this project demonstrates that the rPPG signal can effectively be decoded from the beam profile of structured light NIR illumination sources—an entirely novel finding that has not been examined by any prior research.

Furthermore, the findings of this project demonstrate that motion poses a more significant noise factor than heterogeneous illumination for rPPG in DMS equipped with a narrow-band bandpass filter about 940 nm. The results also suggest that makeup is not a significant noise factor for NIR rPPG performance, that existing architectures generalise poorly to subjects with heart rates significantly outside of the range of their training data, and that spoof detection is feasible using existing state-of-the-art neural rPPG architectures.

Though the RVM camera configuration was initially expected to result in a significant reduction in rPPG performance, the results of this project show that this degradation is actually minimal and effective rPPG sensing is possible from the RVM camera configuration. These results also suggest that rPPG may be performed in OMS from a single camera configuration.

These initial findings establish a solid starting point for future research, with the GitHub repository that houses the programmatic infrastructure developed throughout this project and the new data collection protocols and systems setting the stage for further advancements in rPPG technology at Seeing Machines once a properly synchronised and sufficiently large dataset is acquired.

A. Privacy Concerns

Incorporation of physiological sensing has the potential to greatly enhance the capabilities of existing DMS. If done properly, this could result in a significant reduction in injuries and fatalities as a result of distraction, fatigue, and intoxication. However, it also raises serious privacy concerns, as these methods could be used to measure personal physiological information without the subject's knowledge.

To address these concerns, it is crucial to ensure transparency when using non-contact physiological sensing methods. Just as with traditional contact sensors, individuals must be informed and provide explicit consent before their physiological data is measured or recorded. Furthermore, there should be no penalty for those who choose not to participate. The ability to measure physiological signals in more contexts does not inherently justify doing so without proper regulation. In fact, as the reach of these technologies expands, stricter regulations should be applied, not lessened.

To mitigate such risks, the findings and methodologies developed in this project are protected under Seeing Machines' strict data governance policies, ensuring that only authorised personnel have access to this sensitive data. Additionally, future deployments of this technology must comply with industry-standard data privacy laws and regulations, such as the Australian Privacy Act 1988, to ensure that users' rights are protected.

B. Future Works

Moving forward, expanding the framework developed in this project to include more diverse architectures, proper fusion with depth estimation, and extending models to include respiration sensing will provide greater insights into the value of depth data for three-dimensional sensing.

The training process for this project did not involve data augmentation, which is traditionally a straightforward way to enrich the training dataset. While augmenting physiological signals presents unique challenges, prior studies, such as the augmentation techniques proposed by [54], have demonstrated its feasibility. In particular, the addition of motion augmentation has been shown to be a powerful tool in boosting the performance of neural rPPG methods [55]. Integrating similar augmentation strategies could provide a significant boost to model performance and should be considered in future research.

Configurations for the outdoor dataset have been established to further validate the model under real-world conditions. These scenarios will include static subjects in a car park environment with varying levels and frequencies of ambient illumination. This campaign aims to provide deeper insights into each model's handling of heterogeneous illumination in conjunction with subject movement.

Despite the challenges posed by varying illumination conditions, with a sufficiently large dataset, it is anticipated that the model can be trained to manage these variations as long as they remain below the Nyquist rate of the camera's sampling frequency. To handle more extreme fluctuations, future models may need to incorporate predictive capabilities, generating extended forecasts of the PPG signal from shorter video segments. This approach would resemble an observer mechanism in control systems, enabling the algorithm to maintain robust predictions even when confronted with noisy image data.

Existing Seeing Machines OMS and DMS systems already include and budget for the compute time needed for face tracking. Future research should investigate whether cropping of the input data based on face landmarks can be used to boost performance. Furthermore, the results of this project found that the RVM camera configuration does not pose a significant challenge to existing rPPG neural methods. This suggests that rPPG sensing could be applied in OMS. Future datasets should include the addition of occupants in vehicle to assess this capability.

Finally, the findings of this project suggest more aggressive spatial downsampling may be beneficial for flood-based rPPG systems, and that higher resolution inputs could be beneficial for dot-based rPPG systems in obtaining a more rich set of features from the dot beam profiles. These should also be investigated by future research.

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